

3-Phase Motor-Control Shield for NUCLEO-G474RE

Compatible host module: NUCLEO-G474RE

Document: BOM-KE-G474-SHIELD-A

Revision: A

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Prepared by: Kestrel Electronics Pvt Ltd, Plant 2, Hosur

Internal engineering reference document. NUCLEO-G474RE is named solely as the third-party host module this shield is designed to plug onto; this is not an STMicroelectronics publication.

1. Features

- Plugs directly onto the Arduino Uno V3 and ST morpho headers of a NUCLEO-G474RE (host module supplied separately; target device STM32G474RET6).
- Three-phase gate driver with integrated current-shunt amplifiers, sized for field-oriented control loops running on the host module's HRTIM and comparator peripherals.
- Six N-channel power MOSFETs, 100 V / 180 A per device, two per phase leg.
- Isolated CAN transceiver for multi-axis or vehicle-network integration, independent of the host module's own USB/UART debug path.
- On-board 3.3 V logic rail and a separate buck-regulated gate-drive rail, so the shield never loads the host module's own supply.
- ESD-protected auxiliary USB port and an Ethernet PHY footprint for networked deployments.
- Screw-terminal DC-bus and motor-phase connections, keyed against reversed mating with the host header set.
- Optional Wi-Fi telemetry module footprint (DNP on the standard build -- see BOM note 1).

2. Description

This shield turns a NUCLEO-G474RE into a self-contained three-phase motor-drive evaluation platform. The host module contributes the STM32G474RET6 microcontroller, its ST-LINK debugger, and the timer and ADC resources the STM32 Motor Control SDK's field-oriented control examples expect; the shield contributes everything the motor itself needs -- gate drive, power switching, current sensing, and the DC-bus and phase connections -- so none of it has to be routed through, or draw current from, the host module's own circuitry.

Keeping the power stage on a separate board from the microcontroller module is deliberate: a fault on the motor side (a shorted phase, a reversed DC-bus connection) stays confined to the shield's own regulators and MOSFETs rather than reaching back into the host module's supply rails or debug interface.

3. Host Module

This shield does not include a microcontroller of its own. All firmware runs on the NUCLEO-G474RE it is seated on; the facts below are reference information about that host module, not part of this document's bill of materials.

Host module	NUCLEO-G474RE
Target device	STM32G474RET6
Package	LQFP64
On-board debugger	STLINK-V3E
High-speed external oscillator	24 MHz
Expansion connectors	Arduino Uno V3 + ST morpho
Debug connector	MIPI-10

4. Power Architecture

The shield derives its own 3.3 V logic supply from U3, an LDO fed from the DC bus rather than from the host module's 5 V rail, so the gate driver and CAN transceiver stay powered even while the host module is connected only over USB for debugging with no DC bus present.

A separate buck stage (U2) regulates the gate-drive rail the driver IC (U1) switches from, sized for the six MOSFETs' total gate charge at the shield's rated switching frequency. The two rails share a common ground with the host module across the connector pair (J1, J2), but neither supply is derived from the other.

5. Development Environment

- STM32CubeIDE, with the STM32 Motor Control SDK's field-oriented control workbench targeting the host module's peripheral map.
- IAR Embedded Workbench and Keil MDK-ARM are both supported for the host module; neither toolchain needs shield-specific configuration beyond the pin assignments in the schematic.
- The host module's on-board ST-LINK provides programming and debug over the same USB connection used for power during bring-up (DC bus disconnected).

6. Assembly and Bring-Up

- Confirm the DC bus is disconnected before seating the shield on the host module's headers.
- Seat the shield fully on both the Arduino Uno V3 and ST morpho headers -- a partial seat on only one connector is the most common assembly fault and can expose host module I/O to gate-drive voltages.
- Apply DC bus power only after the host module reports a successful ST-LINK enumeration over USB.
- Load the Motor Control SDK workbench project for this shield's motor profile before the first spin-up; the default HRTIM configuration assumes this shield's gate-drive rail voltage, not a bare host module.

7. Ordering Information

Order code	KE-G474-SHIELD-A
Compatible host module	NUCLEO-G474RE
Also fits	Any Nucleo-64 board sharing the Arduino Uno V3 + ST morpho footprint and a 3.3 V I/O rail
Not compatible with	Nucleo-32 or Nucleo-144 board footprints

8. Connector Pinout

J1 and J2 are the shield's own screw-terminal and header connections -- not the Arduino Uno V3 / ST morpho footprint underneath the board, which is not repeated here.

J1 pin 1-2	DC bus +, DC bus return
J1 pin 3-8	Motor phases A, B, C (two pins each, for crimp strain relief)
J2 pin 1-2	Shield 3.3 V rail out, ground (for probing only -- not a supply input)
J2 pin 3-4	CAN_H, CAN_L
J2 pin 5-6	Aux USB D+, D-
J2 pin 7-8	Ethernet PHY RMII reference clock, ground

9. Test Points and Debug Access

- TP1 -- gate-drive rail output (U2), referenced to shield ground.
- TP2 -- 3.3 V logic rail output (U3).
- TP3-TP5 -- per-phase current-shunt amplifier outputs, direct from U1, before any host-module ADC scaling.
- TP6 -- CAN_H/CAN_L differential pair, for bus-loading measurements with the shield isolated from the host module's own debug session.

10. Safety and Compliance

- The DC bus connector (J1) can carry motor-drive voltages well above the host module's own 3.3-5 V logic levels. Disconnect the DC bus before seating or removing the shield.
- U1, U2 and the MOSFETs (Q1-Q6) are ESD sensitive; handle per ANSI/ESD S20.20 outside of the assembled unit.
- All parts are RoHS 3 (EU 2015/863) and REACH compliant per the manufacturer declarations on file; see BOM note 3.
- This shield is an engineering evaluation aid, not a certified end product -- it carries no independent safety agency marking of its own beyond the individual components' certifications.

11. Firmware and Example Projects

- Field-oriented control: the STM32 Motor Control SDK's FOC workbench, reconfigured for this shield's gate-drive rail and current-shunt scaling (see section 4).
- CAN telemetry: a minimal periodic-broadcast example over U4/U5, useful as a starting point before integrating a project-specific message set.
- Network bring-up: a DHCP-and-ping example exercising U8's RMII path, to confirm the Ethernet PHY footprint before building on it.
- None of these examples are included in this document -- they are distributed with the Motor Control SDK and referenced here for orientation only.

The bill of materials for this shield begins on the next page. Board design resources for the host module itself (schematics, EDA files) are published separately by STMicroelectronics.

ASSEMBLY P/N	DESCRIPTION	REVISION	DOCUMENT	RELEASED	CHANGE REF
KE-G474-SHIELD	3-Phase Motor-Control Shield for NUCLEO-G474RE	A	BOM-KE-G474-SHIELD-A	2026-09-11	ECO-2026-0512
Prepared: R. Nair, Hardware Engineering Checked: S. Iyer, NPI Approved: M. Prasad, Engineering Manager Classification: Internal					

Item	Ref Des	Qty	Manufacturer	Manufacturer P/N	Description	Package	Lifecycle	RoHS	DNP	Approved Alternate
1	U1	1	Texas Instruments	DRV8301DCAR	Three-phase gate driver with buck an	HTSSOP-56	ACTIVE	Yes		
2	Q1-Q6	6	Infineon	IRFB4110PBF	N-channel MOSFET 100 V 180 A, TO	TO-220AB	ACTIVE	Yes		
3	U2	1	Texas Instruments	TPS54331DDAR	3 A step-down converter, 3.5-28 V in	SOIC-8 Pow	ACTIVE	Yes		
4	U3	1	Advanced Monolithic	AMS1117-3.3	1 A LDO, fixed 3.3 V, SOT-223	SOT-223	ACTIVE	Yes		NCP1117ST33T3G
5	U4	1	Texas Instruments	TCAN332DR	3.3 V CAN transceiver, 1 Mbps, SOIC	SOIC-8	ACTIVE	Yes		SN65HVD230DR
6	U5	1	Microchip	MCP2551-I/SN	5 V CAN transceiver, 1 Mbps, SOIC-8	SOIC-8	ACTIVE	Yes		
7	U6	1	Winbond	W25Q128JVSQ	128 Mbit SPI NOR flash, SOIC-8 208	SOIC-8 208r	ACTIVE	Yes		MX25L12835FM2I-10G
8	U7	1	STMicroelectronics	USBLC6-2SC6	Dual-line USB ESD protection, SOT-2	SOT-23-6	ACTIVE	Yes		
9	U8	1	Microchip	KSZ8081RNBIA-TR	10/100 Ethernet PHY with RMII, QFN	QFN-32	ACTIVE	Yes		LAN8720AI-CP-TR
10	Y1	1	Abrakon	ABM8-25.000MHZ-D2Y-7	25 MHz crystal, 18 pF, 20 ppm, SMD	SMD-3225	ACTIVE	Yes		
11	L1	1	Bourns	SRN6045TA-100M	10 uH 3.0 A shielded power inductor,	SMD-6045	ACTIVE	Yes		
12	D1-D2	2	Littelfuse	SMAJ33A	400 W TVS diode, 33 V unidirectional	DO-214AC	ACTIVE	Yes		
13	C1-C16	16	Murata	GRM188R71H104KA93D	100 nF 50 V X7R 10% 0603	0603	ACTIVE	Yes		CL10B104KB8NNNC
14	C17-C24	8	Murata	GRM21BR61E106KA73L	10 uF 25 V X5R 10% 0805	0805	ACTIVE	Yes		
15	R1-R10	10	Yageo	RC0603FR-0710KL	10 kohm 1% 0.1 W 0603	0603	ACTIVE	Yes		
16	R11-R14	4	Yageo	RC0603FR-07100RL	100 ohm 1% 0.1 W 0603	0603	ACTIVE	Yes		
17	R15-R18	4	Yageo	RC0603FR-074K7L	4.7 kohm 1% 0.1 W 0603	0603	ACTIVE	Yes		
18	J1-J2	2	Molex	43045-0400	Micro-Fit 3.0 header, 4 position, ri	Header	ACTIVE	Yes		
19	U9	1	Espressif	ESP32-WROOM-32E-N8	Wi-Fi + BT module, 8 MB flash, SMD	SMD Module	ACTIVE	Yes	DNP	
20	U10	1	Microchip	MCP2515-I/SO	Stand-alone CAN controller with SPI	SOIC-18	NRND	Yes		
21	U11	1	Texas Instruments	LM358BDR	Dual operational amplifier	SOIC-8	ACTIVE	Yes		

Notes

- U9 is shown on the assembly drawing but is DNP on the standard build; populate only for the Wi-Fi telemetry variant.
- Approved alternates are drop-in equivalents released by engineering. Any other substitution requires an approved deviation.
- All parts RoHS 3 (EU 2015/863) compliant. Certificates of conformance required at first article.
- U1 and U6 are MSL 3: bake per J-STD-033 if floor life is exceeded. ESD sensitive, handle per ANSI/ESD S20.20.
- This shield supplies its own logic and gate-drive rails (U2, U3) and does not draw motor-side current from the host module's regulators.
- U10 is NRND. Engineering to confirm a second source before the next build release.
- Mates to the host module's Arduino Uno V3 footprint. Verify J1/J2 keying before power-up -- reversed mating exposes host module I/O to the DC bus.
- Total line items 21. Placements per shield 64, excluding DNP.

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Revision history

Rev	Date	Change ref	Description
A	2026-09-11	ECO-2026-0512	Initial release: shield BOM for the NUCLEO-G474RE motor-control carrier, 21 line items.